IGCSE 0607 Practice Questions — E6.4

Surface area and volume

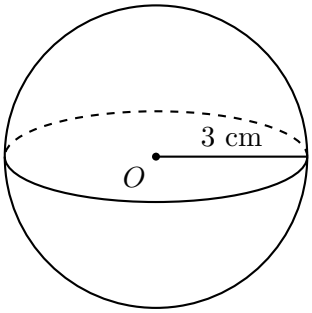
Total Questions: 18 — **Total Marks:** 138 Marks

Table of Contents & Quick Navigation

- **E6.4** Surface area and volume
  → **Paper 2 (Non-Calculator)** *(7 Qs)* · **Paper 4 (Calculator)** *(11 Qs)*

---

1 The diagram shows a solid sphere with center O and radius 3 cm.



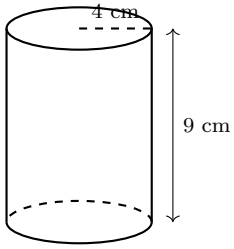
(a) Calculate the exact volume of the sphere.
Give your answer in terms of π.

..... cm^3 **[2]**

(b) Calculate the exact surface area of the sphere.
Give your answer in terms of π.

..... cm^2 **[2]**

2 The diagram shows a solid cylinder of base radius 4 cm and height 9 cm.



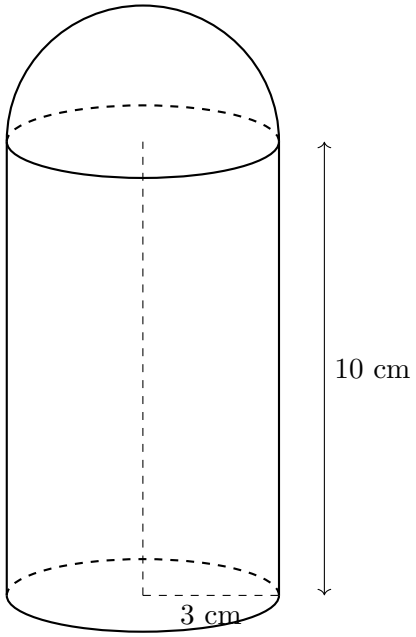
(a) Calculate the volume of the cylinder.
Give your answer in terms of π.

Volume = ... cm^3 **[2]**

(b) Calculate the total surface area of the solid cylinder.
Give your answer in terms of π.

Surface area = ... cm^2 **[3]**

3 A solid is made by joining a solid hemisphere of radius 3 cm to a solid cylinder of radius 3 cm and height 10 cm.



NOT TO SCALE

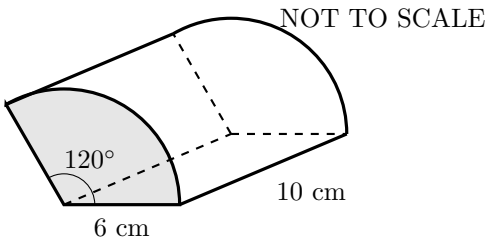
(a) Work out the volume of the solid.
Give your answer in terms of π.

..... cm^3 **[3]**

(b) Work out the total surface area of the solid.
Give your answer in terms of π.

..... cm^2 **[3]**

4 The diagram shows a solid prism of length 10 cm. The uniform cross-section of the prism is a sector of a circle of radius 6 cm and sector angle 120°.



(a) Giving your answers in terms of π, calculate

(i) the area of the sector cross-section,

..... cm^2 **[2]**

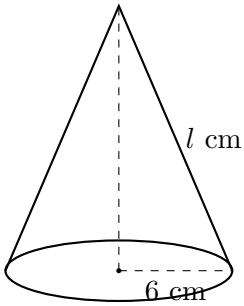
(ii) the volume of the prism.

..... cm^3 **[1]**

(b) Calculate the total surface area of the prism, giving your answer in the form $a + b\pi$, where a and b are integers.

..... cm^2 **[4]**

5 The diagram shows a solid cone.
The base radius of the cone is 6 cm and the slant height is l cm.
The total surface area of the cone is 96π cm^2.



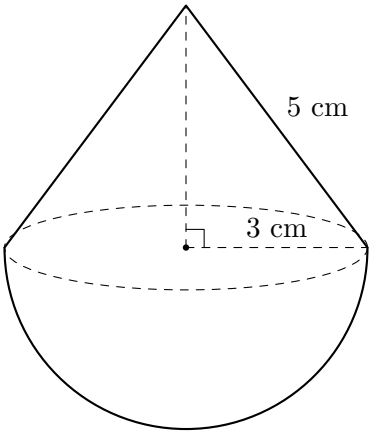
(a) Show that the slant height, l, is 10 cm.

[3]

(b) Work out the volume of the cone.
Give your answer in terms of π.

Volume = ... cm^3 **[4]**

6 The diagram shows a solid formed by joining a cone to a hemisphere with the same radius.
The radius of the hemisphere is 3 cm and the slant height of the cone is 5 cm.



(a) Calculate the exact volume of the solid.
Give your answer in terms of π.

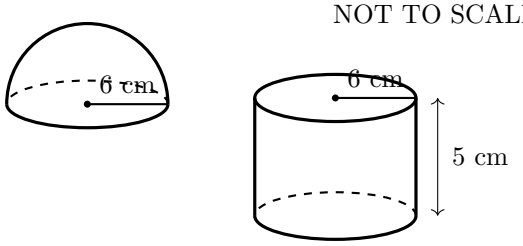
..... cm^3 **[4]**

(b) Calculate the total surface area of the solid.
Give your answer in terms of π.

..... cm^2 **[3]**

7 The diagram shows a solid metal hemisphere of radius 6 cm and a solid metal cylinder of base radius 6 cm and height 5 cm.

NOT TO SCALE



(a) Giving your answers in terms of π, calculate

(i) the volume of the solid hemisphere,

..... cm^3 **[2]**

(ii) the total volume of metal in the hemisphere and the cylinder.

..... cm^3 **[2]**

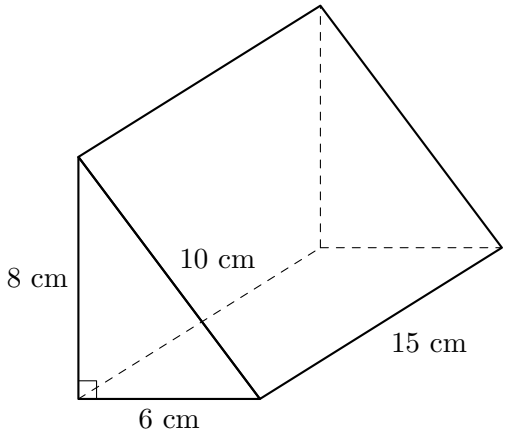
(b) The hemisphere and the cylinder are melted down and recast, with no loss of metal, into a single solid cone of base radius 9 cm. Show that the vertical height of the cone is 12 cm.

[2]

(c) Calculate the total surface area of the solid cone, giving your answer in terms of π.

..... cm^2 **[3]**

8 The diagram shows a solid triangular prism.
The cross-section of the prism is a right-angled triangle with sides 6 cm, 8 cm and 10 cm.
The length of the prism is 15 cm.



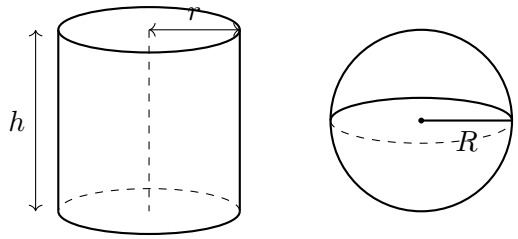
(a) Calculate the volume of the prism.

..... cm^3 **[2]**

(b) Calculate the total surface area of the prism.

..... cm^2 **[3]**

9 The diagram shows a solid cylinder and a solid sphere.



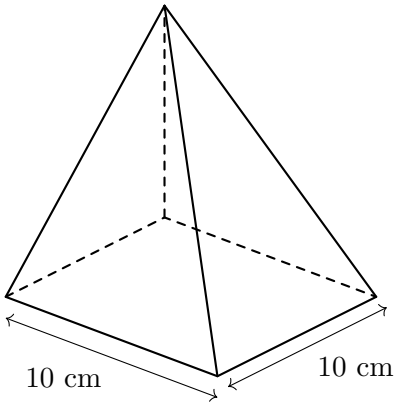
(a) The cylinder has radius $r = 5$ cm and height $h = 12$ cm.
Work out the total surface area of the cylinder.

Surface area = ... cm^2 **[3]**

(b) The sphere has the same volume as the cylinder.
Calculate the radius, R, of the sphere.

$R =$ ..... cm **[3]**

10 The diagram shows a solid pyramid with a square base.
The base has sides of length 10 cm.
The volume of the pyramid is 400 cm^3.



NOT TO SCALE

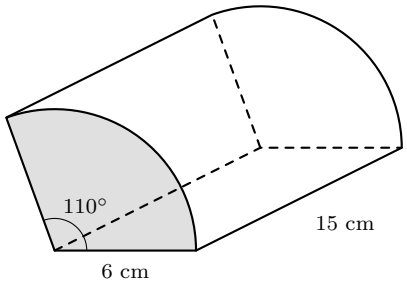
(a) Calculate the perpendicular height of the pyramid.

..... cm **[2]**

(b) Calculate the total surface area of the pyramid.

..... cm^2 **[4]**

11 The diagram shows a prism of length 15 cm.
The cross-section of the prism is a sector of a circle with radius 6 cm and sector angle 110°.



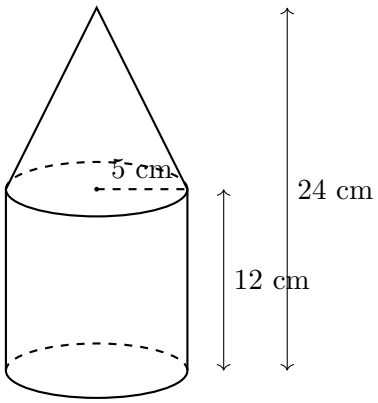
(a) Calculate the volume of the prism.

Volume = ... cm^3 **[3]**

(b) Calculate the total surface area of the prism.

Surface area = ... cm^2 **[3]**

12 A solid is formed by attaching a right circular cone to the top of a solid cylinder.
The cylinder and the base of the cone both have a radius of 5 cm.
The height of the cylinder is 12 cm and the total height of the solid is 24 cm.



(a) Calculate the volume of the solid.

..... cm^3 **[4]**

(b) Calculate the total surface area of the solid.

..... cm^2 **[5]**

13 A solid metal cylinder has radius x cm and height $3x$ cm.
The cylinder is melted down and completely recast into solid spheres.
Each sphere has a radius of $\frac{x}{2}$ cm.

(a) Find an expression, in terms of x and π, for the volume of the cylinder.
Give your answer in its simplest form.

..... cm^3 **[2]**

(b) Find an expression, in terms of x and π, for the volume of one sphere.
Give your answer in its simplest form.

..... cm^3 **[2]**

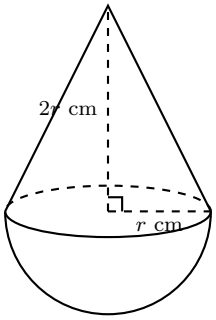
(c) Work out the number of complete spheres that can be made.

..... **[1]**

(d) The total surface area of the original cylinder is 140 cm^2.
Work out the value of x.

$x =$ ..... **[3]**

14 A solid toy is made by joining a solid right circular cone to the flat circular face of a solid hemisphere.
Both the cone and the hemisphere have radius r cm.
The perpendicular height of the cone is $2r$ cm.



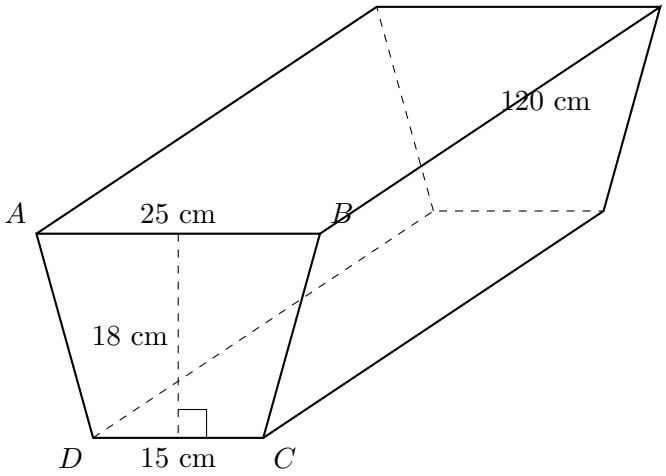
(a) The total volume of the solid toy is 450π cm^3.
Show that $r = 3\sqrt[3]{12.5}$.

[4]

(b) Calculate the total surface area of the solid toy.
Give your answer correct to 3 significant figures.

Surface area = ... cm^2 **[4]**

15 The diagram shows a water trough in the shape of a prism of length 120 cm.
The cross-section of the prism is a trapezium $ABCD$.
AB is parallel to DC.
$AB = 25$ cm, $DC = 15$ cm and the perpendicular distance between AB and DC is 18 cm.



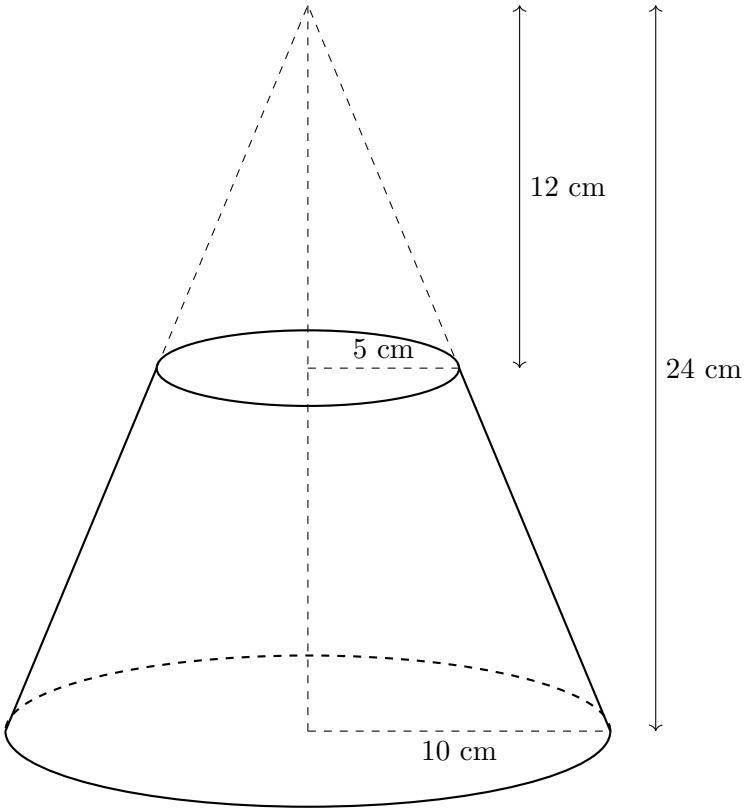
(a) Calculate the volume of the trough in litres.

Volume = ... litres **[4]**

(b) The trough is initially completely empty.
Water flows into the trough at a constant rate of 0.8 litres per second.
Calculate the depth of the water in the trough after 30 seconds.

Depth = ... cm **[5]**

16 A solid is made by removing a small cone from a large cone to form a frustum.
The large cone has a base radius of 10 cm and a perpendicular height of 24 cm.
The small cone has a base radius of 5 cm and a perpendicular height of 12 cm.



NOT TO SCALE

(a) Calculate the volume of the frustum.

..... cm^3 **[3]**

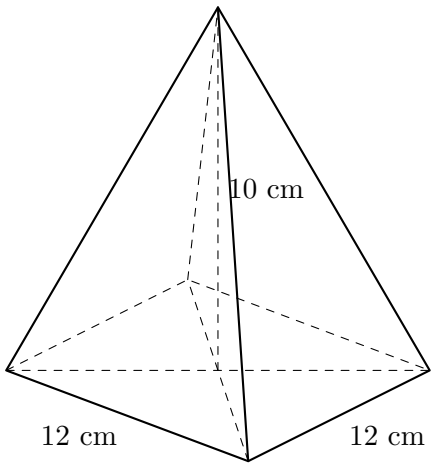
(b) Calculate the total surface area of the solid frustum.

..... cm^2 **[4]**

(c) The solid frustum is melted down to form solid spheres of radius 3 cm.
Calculate the maximum number of complete spheres that can be made.

..... **[3]**

17 A solid metal pyramid has a square base of side 12 cm and a perpendicular height of 10 cm.



(a) Calculate the volume of the pyramid.

..... cm^3 **[2]**

(b) Calculate the total surface area of the pyramid.

..... cm^2 **[4]**

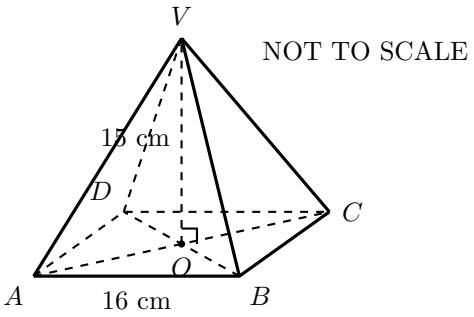
(c) The metal pyramid is melted down and recast to make solid spheres of radius 2.5 cm.
Calculate the number of complete spheres that can be made.

..... **[3]**

(d) A mathematically similar pyramid has a volume of 1620 cm^3.
Calculate the perpendicular height of this similar pyramid.

..... cm **[3]**

18 The diagram shows a solid right pyramid $VABCD$ with a horizontal square base $ABCD$ of side 16 cm.
O is the centre of the base and the vertical height $VO = 15$ cm.



(a) Calculate the volume of pyramid $VABCD$.

..... cm^3 **[2]**

(b) Calculate the total surface area of pyramid $VABCD$.

..... cm^2 **[3]**

(c) The pyramid is cut by a horizontal plane parallel to the base $ABCD$ at a vertical height of 9 cm above the base.
The top small pyramid is removed, leaving a solid frustum.

(i) Calculate the volume of the frustum.

..... cm^3 **[3]**

(ii) Calculate the total surface area of the frustum.

..... cm^2 **[3]**

(d) Calculate the angle that the slant edge VA makes with the horizontal base $ABCD$, giving your answer correct to 1 decimal place.

..... $^\circ$ **[3]**